MA1522 Linear Algebra for Computing
L2 Lesson 13



Tan Kai Meng

Semester 1, 2026/27

Dot/Inner Product

Let $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{R}^n$. Then

$$\boldsymbol{u} \cdot \boldsymbol{v} := u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

Theorem

Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{R}^n$, and $c \in \mathbb{R}$. Then

1. $\boldsymbol{u} \cdot \boldsymbol{v} = \boldsymbol{v} \cdot \boldsymbol{u}$.
2. $(\boldsymbol{u} + \boldsymbol{v}) \cdot \boldsymbol{w} = \boldsymbol{u} \cdot \boldsymbol{w} + \boldsymbol{v} \cdot \boldsymbol{w}$.
3. $(c\boldsymbol{u}) \cdot \boldsymbol{v} = c(\boldsymbol{u} \cdot \boldsymbol{v}) = \boldsymbol{u} \cdot (c\boldsymbol{v})$.
4. $\boldsymbol{v} \cdot \boldsymbol{v} \geq 0$, *with equality if and only if* $\boldsymbol{v} = \boldsymbol{0}$.

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Question 1

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Let $\boldsymbol{a}_1 = (1, 3, -2, 0), \quad \boldsymbol{a}_2 = (2, 6, -5, -2), \quad \boldsymbol{a}_3 = (0, 0, 5, 10).$

By expressing the equations $\boldsymbol{a}_i \cdot \boldsymbol{x} = 0$ for all $i \in \{1, 2, 3\}$ in the form of $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ for some matrix $\boldsymbol{A}$, find all $\boldsymbol{x} \in \mathbb{R}^4$ satisfying

$$\boldsymbol{a}_1 \cdot \boldsymbol{x} = \boldsymbol{a}_2 \cdot \boldsymbol{x} = \boldsymbol{a}_3 \cdot \boldsymbol{x} = 0.$$

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Let $x_2 = s$ and $x_4 = t$ be arbitrary parameters. Then $x_1 = -3s - 4t$ and $x_3 = -2t$.

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Let $x_2 = s$ and $x_4 = t$ be arbitrary parameters. Then $x_1 = -3s - 4t$ and $x_3 = -2t$.

Ans: $\boldsymbol{x} = (-3s - 4t, s, -2t, t) = s(-3, 1, 0, 0) + t(-4, 0, -2, 1),\ s, t \in \mathbb{R}$.

□

Norm of a Vector

Let $\boldsymbol{u} = (u_1, u_2, \ldots, u_n) \in \mathbb{R}^n$.
The **norm** of $\boldsymbol{u}$, denoted $\|\boldsymbol{u}\|$, is

$$\|\boldsymbol{u}\| := \sqrt{\boldsymbol{u} \cdot \boldsymbol{u}} = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}.$$

Theorem

Let $\boldsymbol{u} \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

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Orthogonal Sets and Orthonormal Sets

Let $S = \{\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_k\} \subseteq \mathbb{R}^n$.

S is **orthogonal** if $\boldsymbol{v}_i \cdot \boldsymbol{v}_j = 0$ for all $i \neq j$.

S is **orthonormal** if $\boldsymbol{v}_i \cdot \boldsymbol{v}_j = 0$ for all $i \neq j$ and $\|\boldsymbol{v}_i\| = 1$ for all i.

Clearly,

$$\text{orthonormal} \Rightarrow \text{orthogonal}$$

Theorem

Suppose that S is orthogonal and $\boldsymbol{v}_i \neq \boldsymbol{0}$ for all i. Then

1. *S is linearly independent.*
2. *$\{\frac{1}{\|\boldsymbol{v}_1\|}\boldsymbol{v}_1, \frac{1}{\|\boldsymbol{v}_2\|}\boldsymbol{v}_2, \ldots, \frac{1}{\|\boldsymbol{v}_k\|}\boldsymbol{v}_k\}$ is orthonormal.*

Note

If S is orthogonal, then so is $S \cup \{\boldsymbol{0}\}$.

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Question 2

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Let $S = \{\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_k\} \subseteq \mathbb{R}^n$, and let $\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_k \end{pmatrix}$.

a. How are $\boldsymbol{A}^T\boldsymbol{A}$ and $\boldsymbol{a}_i \cdot \boldsymbol{a}_j$ related?

b. What is the form of $\boldsymbol{A}^T\boldsymbol{A}$ when S is
   i. orthogonal?
   ii. orthonormal?

Question 2: Solution

a

$$\boldsymbol{A}^T\boldsymbol{A} = \begin{pmatrix} \boldsymbol{a}_1^T \\ \boldsymbol{a}_2^T \\ \vdots \\ \boldsymbol{a}_k^T \end{pmatrix} \begin{pmatrix} \boldsymbol{a}_1 & \boldsymbol{a}_2 & \cdots & \boldsymbol{a}_k \end{pmatrix} = \begin{pmatrix} \boldsymbol{a}_1^T\boldsymbol{a}_1 & \boldsymbol{a}_1^T\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_1^T\boldsymbol{a}_k \\ \boldsymbol{a}_2^T\boldsymbol{a}_1 & \boldsymbol{a}_2^T\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_2^T\boldsymbol{a}_k \\ \vdots & \vdots & & \vdots \\ \boldsymbol{a}_k^T\boldsymbol{a}_1 & \boldsymbol{a}_k^T\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_k^T\boldsymbol{a}_k \end{pmatrix}$$

$$= \begin{pmatrix} \boldsymbol{a}_1\cdot\boldsymbol{a}_1 & \boldsymbol{a}_1\cdot\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_1\cdot\boldsymbol{a}_k \\ \boldsymbol{a}_2\cdot\boldsymbol{a}_1 & \boldsymbol{a}_2\cdot\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_2\cdot\boldsymbol{a}_k \\ \vdots & \vdots & & \vdots \\ \boldsymbol{a}_k\cdot\boldsymbol{a}_1 & \boldsymbol{a}_k\cdot\boldsymbol{a}_2 & \cdots & \boldsymbol{a}_k\cdot\boldsymbol{a}_k \end{pmatrix}.$$

Thus $\boldsymbol{a}_i \cdot \boldsymbol{a}_j$ is the (i, j)-entry of $\boldsymbol{A}^T\boldsymbol{A}$.

b $\boldsymbol{A}^T\boldsymbol{A}$ is diagonal if S is orthogonal, and $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}$ if S is orthonormal.

Question 2: Solution

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b $\boldsymbol{A}^T\boldsymbol{A}$ is diagonal if S is orthogonal, and $\boldsymbol{A}^T\boldsymbol{A} = \boldsymbol{I}$ if S is orthonormal.

Angle Between Two Vectors

Let $\boldsymbol{u} = (u_1, u_2, \ldots, u_n)$ and $\boldsymbol{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{R}^n$.
The **angle** θ between $\boldsymbol{u}$ and $\boldsymbol{v}$ is

$$\theta := \cos^{-1} \frac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|}.$$

This yields

$$\boldsymbol{u} \cdot \boldsymbol{v} = \|\boldsymbol{u}\|\|\boldsymbol{v}\| \cos\theta.$$

Remarks

One of the Cauchy-Schwarz inequalities states that

$$|\boldsymbol{u} \cdot \boldsymbol{v}| \leq \|\boldsymbol{u}\|\|\boldsymbol{v}\|.$$

This ensures that $-1 \leq \dfrac{\boldsymbol{u} \cdot \boldsymbol{v}}{\|\boldsymbol{u}\|\|\boldsymbol{v}\|} \leq 1$, so that θ is well-defined.

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Question 3

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Let $\{\boldsymbol{v}_1, \boldsymbol{v}_2, \boldsymbol{v}_3\} \subseteq \mathbb{R}^n$ be orthonormal. Let

$$\boldsymbol{x} = \boldsymbol{v}_1 - 2\boldsymbol{v}_2 - 2\boldsymbol{v}_3 \quad \text{and} \quad \boldsymbol{y} = 2\boldsymbol{v}_1 - 3\boldsymbol{v}_2 + \boldsymbol{v}_3.$$

Determine the angle θ between $\boldsymbol{x}$ and $\boldsymbol{y}$.

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Determine the angle θ between $\boldsymbol{x}$ and $\boldsymbol{y}$.

Solution.

$$\begin{aligned}
\boldsymbol{x} \cdot \boldsymbol{y} &= (\boldsymbol{v}_1 - 2\boldsymbol{v}_2 - 2\boldsymbol{v}_3) \cdot (2\boldsymbol{v}_1 - 3\boldsymbol{v}_2 + \boldsymbol{v}_3) \\
&= 2(\boldsymbol{v}_1 \cdot \boldsymbol{v}_1) + 6(\boldsymbol{v}_2 \cdot \boldsymbol{v}_2) - 2(\boldsymbol{v}_3 \cdot \boldsymbol{v}_3) = 6. \\
\|\boldsymbol{x}\| &= \sqrt{\boldsymbol{x} \cdot \boldsymbol{x}} = \sqrt{(\boldsymbol{v}_1 - 2\boldsymbol{v}_2 - 2\boldsymbol{v}_3) \cdot (\boldsymbol{v}_1 - 2\boldsymbol{v}_2 - 2\boldsymbol{v}_3)} \\
&= \sqrt{1^2 + (-2)^2 + (-2)^2} = 3. \\
\|\boldsymbol{y}\| &= \sqrt{\boldsymbol{y} \cdot \boldsymbol{y}} = \sqrt{2^2 + (-3)^2 + 1^2} = \sqrt{14}. \\
\theta &= \cos^{-1} \frac{\boldsymbol{x} \cdot \boldsymbol{y}}{\|\boldsymbol{x}\|\|\boldsymbol{y}\|} = \cos^{-1} \frac{2}{\sqrt{14}} \approx 1.00685 \approx 57.69^\circ.
\end{aligned}$$

□

Question 4

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Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$. Which of the following statements is/are true?

a. $\|\boldsymbol{x} + \boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

b. $\|\boldsymbol{x} - \boldsymbol{y}\| \geq \|\boldsymbol{x}\| - \|\boldsymbol{y}\|$.



Question 4

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a. $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$. . . . . **Ans: True**

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Question 4: Solution

a. $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

$$\begin{aligned}\|\boldsymbol{x}+\boldsymbol{y}\|^2 &= (\boldsymbol{x}+\boldsymbol{y})\cdot(\boldsymbol{x}+\boldsymbol{y}) \\ &= \boldsymbol{x}\cdot\boldsymbol{x} + 2\boldsymbol{x}\cdot\boldsymbol{y} + \boldsymbol{y}\cdot\boldsymbol{y} \\ &= \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\|\cos\theta + \|\boldsymbol{y}\|^2 \\ &\leq \|\boldsymbol{x}\|^2 + 2\|\boldsymbol{x}\|\|\boldsymbol{y}\| + \|\boldsymbol{y}\|^2 = (\|\boldsymbol{x}\| + \|\boldsymbol{y}\|)^2,\end{aligned}$$

where θ is the angle between $\boldsymbol{x}$ and $\boldsymbol{y}$.

Since both $\|\boldsymbol{x}+\boldsymbol{y}\|$ and $\|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ are non-negative, the inequality is preserved when taking square-roots, so that $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$.

b. $\|\boldsymbol{x}-\boldsymbol{y}\| \geq \|\boldsymbol{x}\| - \|\boldsymbol{y}\|$.

Let $\boldsymbol{w} = \boldsymbol{x}-\boldsymbol{y}$. Then $\boldsymbol{x} = \boldsymbol{w}+\boldsymbol{y}$. By (a),

$$\|\boldsymbol{x}\| = \|\boldsymbol{w}+\boldsymbol{y}\| \leq \|\boldsymbol{w}\| + \|\boldsymbol{y}\|.$$

Thus

$$\|\boldsymbol{x}-\boldsymbol{y}\| = \|\boldsymbol{w}\| \geq \|\boldsymbol{x}\| - \|\boldsymbol{y}\|.$$

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Thus

$$\|\boldsymbol{x}-\boldsymbol{y}\| = \|\boldsymbol{w}\| \geq \|\boldsymbol{x}\| - \|\boldsymbol{y}\|.$$

Next Lesson ...

We will explore **Sections 5.2 and 5.3**.
Please watch/review the video clips for these sections before then.